

■ 3.5.5 Defining Derivatives

You can define the derivative in *Mathematica* of a function f of one argument simply by an assignment like $f'[x_] = fp[x]$.

This defines the derivative of $f(x)$ to be $fp(x)$. In this case, you could have used $=$ instead of $:=$.

```
In[1]:= f'[x_] := fp[x]
```

The rule for $f'[x_]$ is used to evaluate this derivative.

```
In[2]:= D[f[x^2], x]
Out[2]= 2 x fp[x^2]
```

Differentiating again gives derivatives of fp .

```
In[3]:= D[%, x]
Out[3]= 2 fp[x^2] + 4 x^2 fp'[x^2]
```

This defines a value for the derivative of g at the origin.

```
In[4]:= g'[0] = g0
Out[4]= g0
```

The value for the $g'[0]$ is used.

```
In[5]:= D[g[x]^2, x] /. x->0
Out[5]= 2 g0 g[0]
```

This defines the second derivative of g , with any argument.

```
In[6]:= g''[x_] = gpp[x]
Out[6]= gpp[x]
```

The value defined for the second derivative is used.

```
In[7]:= D[g[x]^2, {x, 2}]
Out[7]= 2 g[x] gpp[x] + 2 g'[x]^2
```

To define derivatives of functions with several arguments, you have to use the general representation of derivatives in *Mathematica*.

$f'[x] := rhs$	define the first derivative of f
$\text{Derivative}[n][f][x] := rhs$	define the n^{th} derivative of f
$\text{Derivative}[n_1, n_2, \dots][g][x_1, x_2, \dots] := rhs$	define derivatives of g with respect to various arguments

Defining derivatives.

This defines the second derivative of g with respect to its second argument. `In[8]:= Derivative[0, 2][g][x-, y-] := g2p[x, y]`